

Tutorial - Probability

Indian Institute of Technology
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Random Variables

Random Process: An Experiment which results in different outcomes each time it is repeated in a similar manner.

Random Variables: A function which assigns a numerical value to each possible outcome of a random experiment.

- ▶ **Discrete Random Variable:** A random variable that can take a countable number of possible values. Eg: let Y be the number of defective items found in a batch of 100.
- ▶ **Continuous Random Variable:** A random variable that can take any value within a continuous range. Eg: The temperature in Chennai at 2 PM — could be 31.2°C , 31.25°C , 31.251°C , etc.



Probability Mass Function (PMF)

Applies to: Discrete Random Variables

Definition: The PMF $P(X = x)$ gives the probability that a discrete random variable X takes the value x .

Properties:

- ▶ $P(X = x) \geq 0$ for all x
- ▶ $\sum_x P(X = x) = 1$

Example: Roll of a fair die: $P(X = x) = \frac{1}{6}$ for $x = 1, 2, 3, 4, 5, 6$



Probability Density Function (PDF)

Applies to: Continuous Random Variables

Definition: The PDF $f(x)$ describes how probability is distributed over values of X .

$$P(a \leq X \leq b) = \int_a^b f(x) dx$$

Properties:

- ▶ $f(x) \geq 0$ for all x
- ▶ $\int_{-\infty}^{\infty} f(x) dx = 1$
- ▶ Probability at a given point is zero

Example: Heights of students $\sim$ Normal distribution:

$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$



Distributions

- ▶ Bernoulli
- ▶ Uniform
- ▶ Binomial
- ▶ Poisson
- ▶ Geometric
- ▶ Normal/ Gaussian
- ▶ Exponential
- ▶ Hypergeometric
- ▶ Negative Binomial

Note to Students: It is recommended to be familiar with the following for each distribution:

- ▶ PMF or PDF expression
- ▶ CDF (Cumulative Distribution Function)
- ▶ Mean and Variance formulas



Problem 1

At least one half of an airplanes engines are required to function in order for it to operate, if each engine functions with probability p , for what values of p is a 4 engine plane more likely to operate than a 2 engine plane ?



Problem 1 - Solution

Model: Each engine functions independently with probability p . Each engine is a *Bernoulli*(p) random variable; the number of working engines in an n -engine plane is $\text{Binomial}(n, p)$.

Operating condition: Plane operates iff at least half of its engines function.

(1) 2-engine plane

Number working $\sim \text{Binomial}(2, p)$. Plane operates if at least 1 engine works.

$$P_2 = P(\text{at least 1}) = 1 - P(0) = 1 - (1 - p)^2 = 2p - p^2.$$

(2) 4-engine plane

Number working $\sim \text{Binomial}(4, p)$. Plane operates if at least 2 engines work.

$$P_4 = \sum_{k=2}^4 \binom{4}{k} p^k (1-p)^{4-k} = 1 - [(1-p)^4 + 4p(1-p)^3].$$

Compare: require $P_4 > P_2$.

$$P_4 - P_2 > 0.$$

Substitute the expressions and simplify:

$$P_4 - P_2 = 1 - [(1-p)^4 + 4p(1-p)^3] - (2p - p^2).$$

After algebraic simplification this factors as

$$P_4 - P_2 = p(1-p)^2(3p-2).$$

Since $p \in [0, 1]$ and $p(1-p)^2 \geq 0$, the sign of $P_4 - P_2$ is determined by $3p - 2$. Thus

$$P_4 > P_2 \iff 3p - 2 > 0 \iff p > \frac{2}{3}.$$

Interpretation: If each engine's reliability p exceeds $2/3$, a 4-engine plane is more likely to operate than a 2-engine plane.



Problem 2

Assuming that the length of the phone calls in minutes is an exponential RV with $\lambda = 0.1$. If someone arrives at a phone booth just before you arrive, find the probability that you will have to wait

- ▶ Less than 5 minutes
- ▶ Between 5 to 10 minutes



Problem 2 - Solution

Given: The length of phone calls (in minutes) follows an *Exponential* random variable with rate parameter $\lambda = 0.1$.

$$f_X(x) = \begin{cases} \lambda e^{-\lambda x}, & x \geq 0, \\ 0, & x < 0. \end{cases}$$

Mean: $E[X] = \frac{1}{\lambda} = 10$ minutes

The person before you has just started their call. Let X = remaining call length.

(a) Probability that you wait less than 5 minutes:

$$P(X < 5) = \int_0^5 0.1e^{-0.1x} dx = \left[-e^{-0.1x} \right]_0^5 = 1 - e^{-0.5}.$$

$P(X < 5) = 0.3935$

(b) Probability that you wait between 5 and 10 minutes:

$$P(5 \leq X \leq 10) = \int_5^{10} 0.1e^{-0.1x} dx = \left[-e^{-0.1x} \right]_5^{10} = e^{-0.5} - e^{-1}.$$

$P(5 \leq X \leq 10) = 0.2387$

Interpretation: There is about a 39.3% chance you'll wait less than 5 minutes, and about a 23.9% chance your wait is between 5 and 10 minutes.



Problem 3

An automated optical inspection (AOI) machine is scanning a production line of Printed Circuit Boards (PCBs). Based on historical data, the probability that a single PCB is defective (misaligned component or soldering error) is $p = 0.04$ (or 4%). The inspection of each board is an independent event. Let X be the random variable representing the number of PCBs inspected up to and including the first defective one found.

1. What is the probability that the first defective PCB found is exactly the 10th board inspected?
2. What is the probability that the machine inspects more than 20 boards before finding the first defect?
3. On average, how many boards must the QA engineer expect to inspect to find a single defect?



Problem 3 - Solution

The random variable X follows a Geometric Distribution, denoted as $X \sim \text{Geo}(p)$, with probability mass function:

$$P(X = k) = (1 - p)^{k-1}p$$

Where:

$p = 0.04$ (Probability of "success", i.e., finding a defect)

$1 - p = 0.96$ (Probability of a non-defective board)

k = The trial number of the first success

2. Solving Part A: The 10th Board We need to find the probability that the first 9 boards are good, and the 10th is defective. Here, $k = 10$.

$$\begin{aligned}P(X = 10) &= (1 - 0.04)^{10-1}(0.04) \\&= (0.96)^9(0.04) \\&\approx 0.6925 \times 0.04 \\&= \mathbf{0.0277} \quad (\approx 2.77\%)\end{aligned}$$

3. Solving Part B: More than 20 Boards ($P(X > 20)$) $P(X > k) = (1 - p)^k$.

$$\begin{aligned}P(X > 20) &= \sum_{k=21}^{\infty} (1 - p)^{k-1}p \\&= (1 - p)^{20} \\&= (0.96)^{20} \\&\approx \mathbf{0.442} \quad (\approx 44.2\%)\end{aligned}$$

Solving Part C: Expected Value ($E[X]$) The expected value (mean) of a geometric distribution represents the average "waiting time" for the first success.

$$E[X] = \frac{1}{p} = \frac{1}{0.04} = 25$$



Questions

- ▶ A communication system transmits the digits 0 and 1. Due to noise, the digit is incorrectly received with probability 0.2. Suppose that repetition coding is used to reduce the error, in which five 1's are transmitted in place of 1 and five 0's are transmitted in place of 0. Majority decoding is used at the receiver. What is the probability that the message will be incorrectly decoded?
- ▶ Let X be a random variable exponentially distributed with parameter $\lambda > 0$. If $5E(X) = \text{Var}(X)$, where $E(X)$ and $\text{Var}(X)$ denote the expectation and variance of X , find the value of λ .
- ▶ Compare the Poisson approximation with the correct Binomial probability for the following cases:
 - (a) $P(X = 2)$ when $n = 10$, $p = 0.1$
 - (b) $P(X = 0)$ when $n = 10$, $p = 0.1$
 - (c) $P(X = 4)$ when $n = 9$, $p = 0.2$

For more on the Poisson approximation to the Binomial distribution, refer to: [Poisson Approximation to the Binomial Distribution](#).



Thanks

